

Introduction to B-Tree & B+ Tree — Data Structures

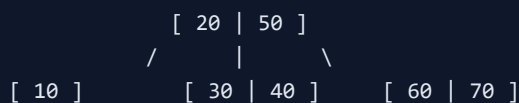
Definition: A **B-Tree** is a self-balancing multi-way search tree of order m where each node can contain multiple keys and more than two children, designed specifically to optimize **disk read/write block accesses** in database indexing and file systems.

1. Detailed Technical Explanation

1. Properties of a B-Tree of Order m :

1. Every node has at most m children.
2. Every internal node (except root) has at least $\lceil m/2 \rceil$ children.
3. The root has at least 2 children if it is not a leaf node.
4. A non-leaf node with k children contains exactly $k - 1$ sorted keys.
5. **All leaves appear on the same physical level** (perfectly balanced).

Sample B-Tree of Order 3:



2. B-Tree vs B+ Tree Comparison

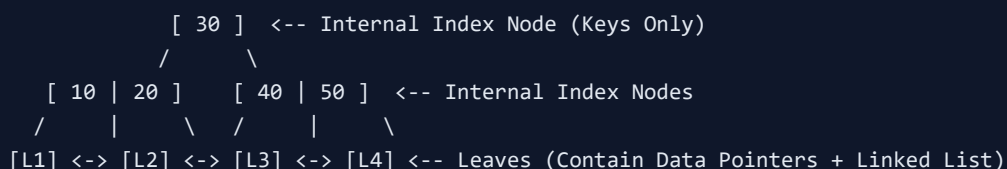
B-TREE:

- Keys AND satellite record data pointers are stored in BOTH internal and leaf nodes.

B+ TREE:

- Internal nodes store ONLY routing search keys.
- ALL satellite data records/pointers reside ONLY in LEAF nodes.
- All leaf nodes are linked together as a SLL / DLL (Doubly Linked List) for sequential range scans!

B+ Tree Structure:



Detailed Comparison Table:

Feature	B-Tree	B+ Tree
Data Storage	Data stored in internal & leaf nodes.	Data stored ONLY in leaf nodes.
Search Efficiency	Search may finish early at internal nodes.	Search always goes down to leaf nodes ($O(\log N)$ uniform).
Range Queries	Inefficient (requires in-order tree traversal).	Extremely Fast (traverse sequential linked list of leaves).
Node Capacity	Fewer keys per disk block due to data pointers.	More keys per block (higher fanout, smaller tree height).
Applications	File systems, database core storage.	Relational Database Indexing (MySQL InnoDB, Oracle, Db2).

3. Core Concepts & Memory Keywords

- **Order m:** Maximum number of children a node can have.
- **Fanout:** Number of branch pointers per index node (high fanout reduces disk I/O).
- **Sequential Leaf Chaining:** Doubly linked leaf nodes enabling $O(K)$ range queries (`WHERE age BETWEEN 20 AND 30`).

4. Must-Write Points for Exams

- B-Trees and B+ Trees reduce disk I/O by fitting large numbers of keys inside a single disk page block (e.g., 4KB or 8KB).
- B+ Trees are preferred over B-Trees for database indexes because leaf chaining enables fast range searches and higher fanout.

5. Quick Recall Flow

Multi-Way Search Tree -> B-Tree (Data in all nodes) vs B+ Tree (Data only in linked leaves + High Fanout for DB Indexing)